

JAEUK SHIN

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google scholar

personal website

linkedin

EDUCATION

B.S., Mathematics, Seoul National University	2014-2018
M.S., Mathematics, Seoul National University	2018-2020
Ph.D., Electrical and Computer Engineering, Seoul National University	2020-2026

WORK EXPERIENCE

Research Engineer Naver Labs , Robotics & Embodied AI Team	Sep 2026 - Present Gyeonggi, South Korea
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- Development of learning-based systems for outdoor mobile robots, spanning data collection, end-to-end model training, and deployment on physical testbeds

PUBLICATIONS

* indicates equal contribution.

Preprints

- **From Prediction Uncertainty to Conformalized Distance Fields for Safe Motion Planning**
Jaeuk Shin*, Yoonseok Ra*, and Insoon Yang
2026.

Journal Publications

- **CANVAS: Competency-Aware Navigation Assessment Suite for Pedestrian Trajectory Forecasting**
Jaeuk Shin, Joon Ho Han, Gihwan Kim, Jungjin Lee, and Insoon Yang
IEEE Open Journal of Control Systems (OJ-CSYS), Early Access, 2026.
- **Multiparametric Analysis of Multi-Task Markov Decision Processes: Structure, Invariance, and Reducibility**
Jaeuk Shin and Insoon Yang
IEEE Control System Letters (L-CSS), vol. 8, pp. 928-933, 2024. (Selected for presentation at CDC'24)
- **Anderson acceleration for partially observable Markov decision processes: A maximum entropy approach**
Mingyu Park*, Jaeuk Shin*, and Insoon Yang
Automatica, vol. 164, 2024.
- **Control of Fab Lifters via Deep Reinforcement Learning: A Semi-MDP Approach**
Giho Kim*, Jaeuk Shin*, Gihun Kim, Joonrak Kim, and Insoon Yang
IEEE Transactions on Automation Science and Engineering (T-ASE), vol. 21, issue 4, pp. 5136-5148, 2024.
- **Infusing model predictive control into meta-reinforcement learning for mobile robots in dynamic environments**
Jaeuk Shin, Astghik Hakobyan, Mingyu Park, Yeoneung Kim, Gihun Kim, and Insoon Yang
IEEE Robotics and Automation Letters (RA-L), vol. 7, issue 4, pp. 10065-10072, 2022. (Selected for presentation at IROS'22)

- **Hamilton-Jacobi deep Q-learning for deterministic continuous-time systems with Lipschitz continuous controls**
Jeongho Kim, **Jaeuk Shin**, and Insoon Yang
Journal of Machine Learning Research (JMLR), vol. 22, no. 206, pp. 1-34, 2021.

Conference Publications

- **KoopCast: Trajectory Forecasting via Koopman Operators**
Jungjin Lee, **Jaeuk Shin**, Gihwan Kim, Joonho Han, and Insoon Yang
IEEE International Conference on Robotics & Automation (ICRA), 2026.
- **Egocentric Conformal Prediction for Safe and Efficient Navigation in Dynamic Cluttered Environments**
Jaeuk Shin, Jungjin Lee, and Insoon Yang
IEEE Conference on Decision and Control (CDC), 2025.
- **On task-relevant loss functions in meta-reinforcement learning**
Jaeuk Shin, Giho Kim, Howon Lee, Joonho Han, and Insoon Yang
Annual Learning for Dynamics & Control Conference (L4DC), 2024.

RESEARCH EXPERIENCE

Research Intern

December 2025 - June 2026

Done in **Naver Labs**

Gyeonggi, South Korea

- Development of an RL-based visuomotor policy for end-to-end autonomous urban navigation, integrating visual perception with control in a mobile robotics platform

Research Collaborator

December 2020 - September 2021

Joint project with **SK hynix**

Seoul, South Korea

- Design of an efficient scheduling algorithm of FAB lifters using deep reinforcement learning

ACADEMIC SERVICES

Reviewer for Journals

- *IEEE Transactions on Robotics (T-RO)*
- *IEEE Transactions on Automation Science and Engineering (T-ASE)*
- *IEEE Robotics and Automation Letters (RA-L)*
- *IEEE Control System Letters (L-CSS)*

Reviewer for Conferences

- *International Conference on Learning Representations (ICLR)*
- *IEEE International Conference on Robotics and Automation (ICRA)*
- *Conference on Robot Learning (CoRL)*
- *ACM International Conference on Hybrid Systems: Computation and Control (HSCC)*
- *American Control Conference (ACC)*
- *IEEE Conference on Decision and Control (CDC)*
- *Annual Learning for Dynamics and Control Conference (L4DC)*

TEACHING EXPERIENCE

Graduate Teaching Assistant

- Undergraduate Calculus II
- Linear Algebra
- Mathematical Theory and Algorithms for Optimization
- Advanced Control Techniques
- Stochastic Control and Reinforcement Learning
- Stochastic Control and Reinforcement Learning
- Advanced Control Techniques
- Linear Algebra

Fall 2018, SNU
Spring 2019, SNU
Fall 2019, SNU
Fall 2020, SNU
Spring 2021, SNU
Spring 2021, SNU
Fall 2022, SNU
Fall 2024, SNU

Teaching Assistant
Samsung Electronics

August 2020, July 2021
Seoul, South Korea

- Deep reinforcement learning lecture in Samsung AI Expert Educational Program

TECHNICAL STRENGTHS

Computer Languages Python, MATLAB, C++